

3.3 Convergence Tests (Divergence Test)

Theorem:

Proof:

Corollary (Test for Divergence):

Proof:

This is simply the contrapositive of the above theorem.

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Example 1: Prove that the series $\sum_{n=1}^{\infty} \left(\frac{n^2}{5n^2 + 4} \right)$ diverges.

Example 2 (you try): In no more than two sentences explain what the test for divergence tells you about the series $\sum_{k=1}^{\infty} \left(\frac{1}{3k} \right)$.

Theorem: If $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent series, then

Proof: The proof follows from the limit laws of section 3.1.1.

Example 3: Determine (with justification) whether the series is convergent or divergent. If it is convergent, find its sum: $\frac{1}{3} + \frac{2}{7} + \frac{1}{27} + \frac{2}{49} + \frac{1}{243} + \frac{2}{343} + \cdots$.

Theorem:

Example 4: Determine whether the following series converges or diverges. $\sum_{i=10}^{\infty} \frac{1}{i} = \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \dots$

Example 5 (you try):

- (a) Determine (with justification) whether the series is convergent or divergent. If it is convergent, find its sum: $\sum_{n=1}^{\infty} \left(\frac{3}{n(n+1)} + \frac{1}{2^n} \right)$.
- (b) Determine (with justification) whether the series is convergent or divergent. If it is convergent, find its sum: $\sum_{n=1}^{\infty} (\arctan(-n))$.
- (c) TRUE or FALSE: If $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are divergent series, then $\sum_{n=1}^{\infty} (a_n + b_n)$ is a divergent series. If you answer TRUE you must explain why this is TRUE. If you answer FALSE you must provide a counterexample proving the statement FALSE.